

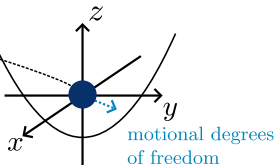
Topological phonons in Rydberg-atom arrays

Christian Gommeringer;

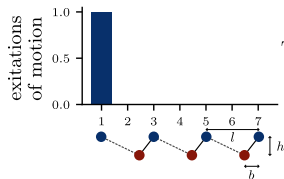
supervisors: Beatriz Olmos

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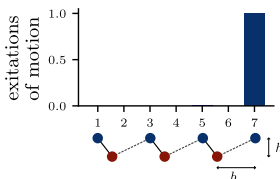
Rydberg atom
in optical tweezer



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Transfer
→

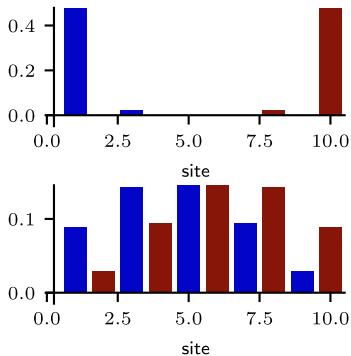
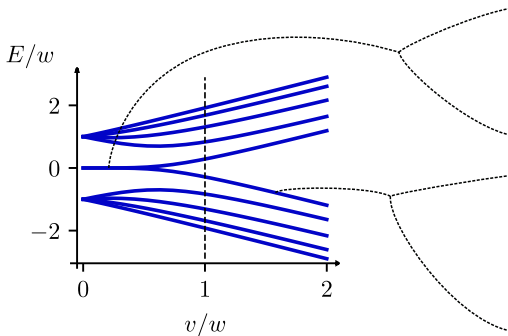
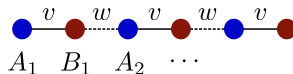


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SSH-model

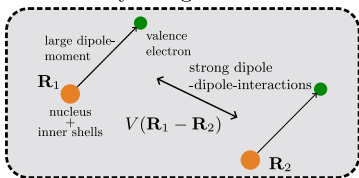
Hamiltonian:

$$H_{\text{ssh}} = \sum_{i=1}^{N/2} \left[v c_{A_i}^\dagger c_{B_i} + w c_{B_i}^\dagger c_{A_{i+1}} \right] + \text{h.c.}$$



Physical implementation

Rydberg atoms



$$H_{\text{eff}} = \sum_{\substack{i,j=1 \\ c,d \in \{a,b\}}}^{N/2} U_{i,j}^{c,d} c_i^\dagger d_j \quad \text{for } i \neq j: U_{ij}^{a,b} = \frac{1}{m\omega} \frac{\partial^2 V(\mathbf{R}_i^A - \mathbf{R}_j^B)}{\partial \delta \mathbf{R}_i^A \partial \delta \mathbf{R}_j^B}$$

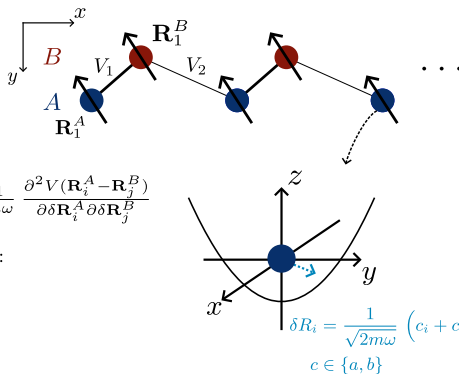
imposing **chiral symmetry**:
sublattices (●,●) can not interact with each other

$$H_{\text{eff}} = \sum_{i,j} U_{i,j}^{a,b} a_i^\dagger b_j + \text{h.c.}$$

1D phonons:

can be mapped to the extended ssh model

preparing the atoms in the Rydberg states:

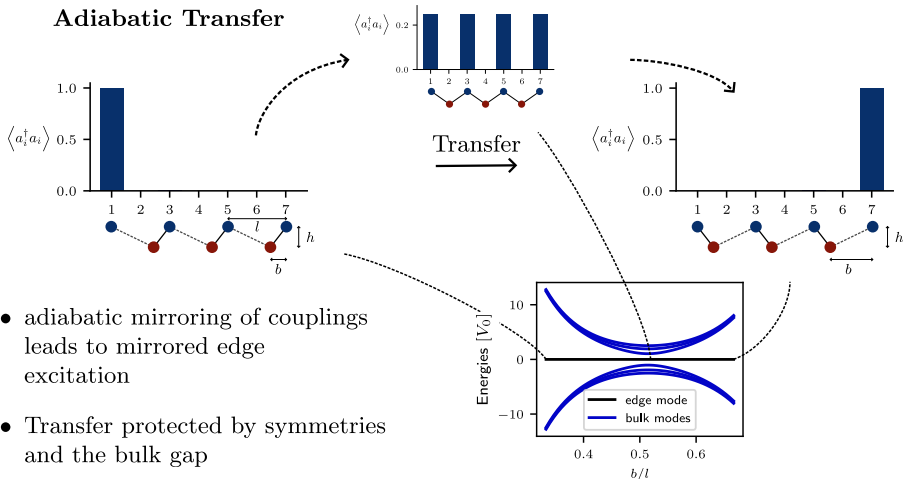


2D phonons:

two independent nearest neighbor ssh models

Quantum state transfer

Adiabatic Transfer



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¹S. Raupach et al., *Robust topological quantum state transfer with long-range interactions in rydberg arrays*, Jan. 27, 2026

Summary and Outlook

Summary

- topological phases for motion in Rydberg tweezer arrays
- can be exploited for quantum state transfer

Outlook

- Spin-phonon coupling:
 $H = H_{\text{sp}} + H_{\text{sp-ph}}$
 - What effect has the spin-phonon coupling on transport performance?

Thank you